

HEALTHCONNECT EXPERIENCE LAB

ADVANCED ANALYTICS & DECISION SUPPORT NOTEBOOK

Week 6 | Data Analytics

Analytical progression — WEEK 5 EVIDENCE → SELECT → DEEPEN → VALIDATE → PRIORITIZE → HANDOFF → INTEGRATE → DECIDE → ACT → MONITOR → PREPARE TO TEST

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Notebook Presentation Standard

This English version is the professional master structure for the executable Week 6 notebook. It follows a decision-oriented analytical flow: every analytical question is tied to reproducible evidence, interpretation, business relevance and an explicit decision or testing implication.

Notebook element	Presentation rule
Section heading	Short, decision-oriented Markdown heading.
Analytical question / hypothesis	State before code; no analysis without a defined question.
Code	One objective per cell; concise comments; suppress non-essential output.
Evidence	Show the relevant table, statistic or visual immediately after computation.
Finding	State only what the evidence supports.
Interpretation	Explain analytical meaning without overstating causality.
Business implication	Explain why the finding matters to HealthConnect.
Decision support	Specify the decision, action or test informed by the evidence.
Guardrail	State population, data-quality or interpretation limitations explicitly.

Core rule — One question → one calculation → one piece of evidence → one interpretation → one business implication → one decision.

1. Week 5 → Week 6 Analytical Transition

Week 5 established the initial analytical baseline for HealthConnect appointment attendance. Booking Lead Time emerged as the clearest observed differentiation in No-Show Rate, while Recorded Previous No-Shows provided a useful complementary signal. Reminder Status and Distance to Clinic showed more limited standalone differentiation.

Week 6 advances this work by testing robustness, evaluating combined signals, revalidating the KPI framework, integrating Data Science evidence, and translating validated findings into measurable attendance-support actions.

Week 5 output	Week 6 advancement
Initial EDA and segmentation	Robustness and sensitivity testing
Signals examined mainly separately	Joint-signal evaluation
Initial KPI framework	KPI revalidation and reclassification
Descriptive findings	Evidence-strength and decision-value hierarchy
Initial recommendations	Specific, measurable and testable recommendations
Analytics-only interpretation	Data Analytics ↔ Data Science integration

2. Analytical Scope & Methodology

Element	Validated definition
Dataset	HealthConnect Appointment Data
Total appointments	5,000
Unit of analysis	One appointment record
Primary analytical population	Attended + No-Show

Attendance-observable appointments	4,737
No-Shows	2,423
Attended	2,314
Cancelled	263 — retained in the dataset but excluded from the primary NSR denominator
Binary analytical outcome	No-Show = 1; Attended = 0

No-Show Rate definition — $\text{No-Shows} / (\text{No-Shows} + \text{Attended}) = 2,423 / 4,737 = 51.15\%$. Cancelled appointments are excluded because attendance is no longer observable once an appointment is cancelled.

2.1 Analytical Guardrails

- Appointment record remains the unit of analysis; no patient-level longitudinal history is reconstructed.
- Association does not establish causality. Observational results are not presented as causal effects.
- Statistical significance does not automatically imply decision relevance; magnitude, robustness and actionability are assessed together.
- Missing Distance to Clinic and Waiting Time values are not arbitrarily imputed.
- Recorded patient-history and demographic variables are interpreted cautiously because patient-level inconsistencies were documented.
- Data Science feature importance indicates predictive relevance within a model; it does not establish a causal mechanism.

3. Single Source of Truth & KPI Revalidation

Measure	Week 6 classification	Validated value	Decision role
No-Show Rate	Primary Outcome KPI	51.15%	Monitor attendance performance
Reminder Coverage Rate	Secondary Process KPI	72.68%	Monitor reminder-process execution
NSR by Reminder Status	Diagnostic Comparative Metric	No: 54.63% Yes: 49.86%	Assess observed differentiation
No-Reminder Gap	Supporting Diagnostic Metric	+4.78 pp	Quantify reminder-status difference
NSR by Booking Lead Time	Diagnostic / Prioritisation Metric	By lead-time band	Identify priority profiles
NSR by Previous No-Shows	Diagnostic / Refining Metric	By recorded history	Refine priority profiles

KPI decision — Week 6 does not promote every newly calculated measure to KPI status. A measure becomes a management KPI when it is explicitly linked to a defined decision or operational action.

4. Advanced Analysis — Booking Lead Time

4.1 Analytical Question

Does the Week 5 Booking Lead Time pattern remain meaningful under alternative segmentation, statistical testing and time-period validation?

4.2 Hypothesis

H_1 : No-Show Rates differ materially according to the time between booking and appointment date.

Booking Lead Time	N	No-Show Rate	95% Wilson CI
0–7 days	604	29.47%	25.97–33.23%
8–14 days	574	35.19%	31.40–39.19%
15–30 days	1,265	45.53%	42.81–48.29%
31–45 days	1,187	57.03%	54.20–59.82%
46–60 days	1,107	71.36%	68.63–73.95%

Test / comparison	Result
Spearman	$\rho = 0.2873$; $p \approx 1.08 \times 10^{-90}$
Mann–Whitney	$p \approx 5.30 \times 10^{-87}$
0–30 vs 31–60 days	39.13% vs 63.95%
Observed gap	+24.82 pp
Risk ratio	≈ 1.63 (95% CI ≈ 1.54 – 1.73)
Logistic OR per +10 days	≈ 1.42 ; $p < 0.001$
Sensitivity	Gradient retained under alternative banding and across analysed time periods

FINDING — Booking Lead Time remains the strongest and most robust observed differentiation in No-Show Rate.

BUSINESS IMPLICATION — Appointments with longer booking lead times are the first population to consider for targeted attendance-support evaluation.

GUARDRAIL — The evidence supports prioritisation for testing; it does not prove that reducing lead time would itself cause fewer no-shows.

5. Advanced Analysis — Recorded Previous No-Shows

Question: Does Recorded Previous No-Shows retain useful differentiation after sparse-category regrouping and robustness checks?

Recorded Previous No-Shows	N	No-Show Rate
0	2,745	46.30%
1	1,482	55.87%
2	419	62.05%
3+	91	70.33%

Comparison	Result
0 vs ≥1 previous no-shows	46.30% vs 57.83%
Observed gap	+11.53 pp
Risk ratio	≈ 1.25
Cramér's V	≈ 0.124
Logistic OR per +1 previous no-show	≈ 1.40
Temporal stability	Direction retained across analysed periods

FINDING — Recorded Previous No-Shows provides confirmed complementary differentiation.

DECISION ROLE — Use this variable to refine prioritisation, not to replace Booking Lead Time as the primary signal.

DATA-QUALITY GUARDRAIL — The field is treated as a recorded appointment attribute. Documented longitudinal inconsistencies prevent interpretation as a perfectly reconstructed patient history.

6. Combined Analysis — Lead Time × Previous No-Shows

Objective: assess whether the two signals provide complementary differentiation and whether a complex interaction is required.

Lead Time Priority Group	Previous NS = 0	Previous NS ≥ 1	Total
0–30 days	34.52%	45.62%	39.13%
31–60 days	59.07%	70.52%	63.95%
Total	46.30%	57.83%	51.15%

Model	Result / interpretation
Lead Time only	AIC ≈ 6,167.1
Additive Lead Time + Previous NS	AIC ≈ 6,092.8 — improved descriptive fit
Lead × Previous NS interaction	AIC ≈ 6,094.4
Interaction term	OR ≈ 1.017; p ≈ 0.496 — not supported

FINDING — Booking Lead Time and Recorded Previous No-Shows provide complementary differentiation, but a complex statistical interaction is not supported.

DECISION — Use Lead Time as the primary signal and Previous No-Shows as a refining signal. Do not build a deterministic risk score from this interaction.

7. Contextual Validation & Data Science Integration

7.1 Reminder Status

Measure	Result
Reminder Coverage Rate	72.68%
NSR — No recorded reminder	54.63%
NSR — Reminder recorded	49.86%
Observed gap	+4.78 pp
Interpretation	Limited observed differentiation; association, not causal evidence

Reminder × Lead Time and Reminder × Previous No-Shows interaction tests did not provide sufficient evidence of a substantial interaction. Reminder Status therefore remains a secondary/contextual signal and a candidate for intervention testing.

7.2 Cross-Track Integration Back — Data Analytics ↔ Data Science

Cross-track collaboration was conducted with AYDEN NGNINTEDEM DEMANOU — Data Science Intern, HealthConnect Pod 01. Analytics provided the validated outcome definition, analytical population, KPI logic, prioritised findings and interpretation guardrails. Data Science returned modelling evidence and feature-level insights that were then reviewed and integrated back into the Analytics decision-support framework.

Analytics conclusion before collaboration	Data Science contribution	Integrated interpretation
Booking Lead Time = primary signal	booking_lead_days and derived representations such as booking_lead_days_sq / long_lead_followup were retained as useful modelling features	STRENGTHENED — convergence between Analytics robustness and predictive relevance
Previous No-Shows = complementary signal	previous_no_shows and related history representation retained in the modelling feature space	CONFIRMED — complementary signal, with the Analytics data-quality guardrail retained
Distance = weak standalone association	distance_to_clinic_km showed predictive usefulness in the Data Science model	NUANCED — retain for multivariable evaluation; do not automatically promote to standalone priority
Age Group = no standalone prioritisation	continuous age showed predictive usefulness in modelling	NUANCED — continuous multivariable predictive value does not justify standalone demographic targeting
Reminder = limited differentiation	Operational relevance retained for attendance-support intervention design	TEST — evaluate reminder/confirmation/support rather than claim a causal reminder effect

COLLABORATION EVIDENCE — The collaboration produced a documented analytical change: Booking Lead Time was independently reinforced, Distance and Age interpretations were nuanced, and a new Distance × Reminder hypothesis was identified for targeted analytical validation.

MODEL-INTERPRETATION RULE — Random Forest importance can shift when derived features are added. Final documentation therefore emphasises retained features and cross-track convergence/nuance rather than freezing obsolete importance percentages.

7.3 Cross-Track Hypothesis — Distance × Reminder Status

The Data Science review raised a plausible operational hypothesis: the relationship between distance and no-show outcomes may differ according to reminder status. This is treated as a hypothesis requiring validation, not as an established mechanism.

CROSS-TRACK HYPOTHESIS — Longer-distance appointments may show different No-Show Rate differentiation depending on whether a reminder is recorded.

GUARDRAIL — Even if a descriptive or statistical interaction is observed, the analysis cannot conclude that reminders causally improve travel planning.

8. Evidence Hierarchy & Decision Translation

Level	Signal	Week 6 conclusion	Decision use
PRIMARY	Booking Lead Time	Strengthened & robust	Prioritise the population for testing
SUPPORTING / COMPLEMENTARY	Recorded Previous No-Shows	Confirmed complementary differentiation	Refine prioritisation
SECONDARY / CONTEXTUAL	Reminder Status	Limited observed differentiation	Test interventions
SECONDARY / CONTEXTUAL	Distance to Clinic	Weak standalone; retained for multivariable evaluation	Context / modelling feature
NO STANDALONE PRIORITISATION	Appointment Type, Day, Time, Previous Appointments, Waiting Time, Age Group, Gender	Insufficient standalone differentiation	Do not prioritise in isolation

DECISION SUPPORT STATEMENT — HealthConnect should prioritise attendance-support evaluation around appointments with longer Booking Lead Times, using Recorded Previous No-Shows as a complementary signal where appropriate. Reminder processes remain operationally relevant, but current observational evidence supports targeted testing rather than a causal effectiveness claim.

9. Specific & Measurable Recommendations

Recommendations are framed as measurable operational tests. Where the dataset does not provide an official business target, the success threshold should be agreed with HealthConnect before the intervention begins.

Action	Target population	Owner	Implementation	KPI / baseline	Decision rule
Test proactive attendance support	Longer lead-time appointments; initial focus on 31–60 days	Clinic Operations / Appointment Support	Targeted reminder, confirmation or attendance-support intervention in a controlled design	NSR; 31–60 baseline = 63.95%	Predefine success criterion; compare intervention and comparison groups; scale only if measurable improvement is observed
Refine targeting with Previous No-Shows	Profiles with ≥1 recorded previous no-show	Analytics + Operations	Use as a complementary filter, not a patient label	NSR by priority profile; ≥1 baseline = 57.83%	Retain only if it improves targeting value without unacceptable data-quality risk
Test reminder / confirmation strategy	Prioritised population with appropriate comparison group	Appointment Support	Test timing, confirmation or support before system-wide change	NSR + Reminder Coverage; global NSR 51.15%; coverage 72.68%	Extend, adjust or stop based on predefined evaluation criteria

Principle — No recommendation is presented as a proven solution before testing. Wider deployment depends on measured results within a defined population and evaluation window.

10. Limitations, Risks, Dependencies & Evidence

Category	Documented issue	Control / implication
Data quality	Patient-level inconsistencies in gender, age, age_group and recorded-history variables	Do not reconstruct longitudinal patient histories; interpret history variables cautiously
Missingness	Distance: 90 missing; Waiting Time: 60 missing	No arbitrary imputation; analyse available cases
Method	Observational analysis	Association ≠ causality; recommendations framed as tests
Statistics	Large N can produce small p-values	Assess effect magnitude, robustness and decision value
Data Science	Feature importance depends on model and feature space	Do not infer causality or freeze obsolete RF percentages
Operational	No detailed intervention cost or capacity data	Do not claim unsupported ROI or financial impact
Cross-track	DS evidence used for convergence and nuance	Document exactly what changed after DS review

11. Week 6 Dashboard — Evidence-Based Improvement

- 01 | MONITOR — validated headline KPIs, Attendance-Observable Appointments and Reminder Status explicitly framed as observed association.
- 02 | DIAGNOSE — strengthened Booking Lead Time evidence, Previous No-Shows regrouped to 0 / 1 / 2 / 3+, updated evidence hierarchy and visible decision gaps.
- 03 | PRIORITIZE — 2×2 Lead Time × Previous No-Shows matrix, prioritisation logic, action framework and guardrail against deterministic patient-risk classification.

Dashboard rule — Update only what the new evidence justifies. Prioritise readability, hierarchy and decision value over visual density.

12. Documented Cross-Track Contribution

Required element	Documentation
Partner track	Data Science
Collaborator	AYDEN NGNINTEDEM DEMANOU — Data Science Intern, HealthConnect Pod 01
Dependency	Analytics supplies validated definitions and signals; Data Science evaluates predictive usefulness
Output provided	Cross-Track Analytical Evidence Brief + Analytics KPI Reference + population/outcome definitions + interpretation guardrails
Output received	Modelling feature evidence, derived features and Data Science interpretation
Activity completed	Analytical handoff, cross-track review, feature discussion and Integration Back
What changed	Booking Lead Time strengthened; Distance and Age nuanced; derived features documented; obsolete RF percentages not frozen; Distance × Reminder hypothesis identified for targeted validation

Evidence available	Briefs, collaboration exchanges, Data Science result captures, updated notebook and dashboard
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Integration criterion — This qualifies as integration because the Data Science contribution produced validation, nuance and documented changes to the Analytics interpretation — not merely communication.

13. Week 7 Testing Requirements

- Validate the stability of the proposed prioritisation within the selected testing populations and evaluation windows.
- Measure whether a targeted intervention changes No-Show Rate relative to an appropriate comparison group.
- Evaluate results by priority profile without over-interpreting sparse subgroups.
- Test reminder, confirmation or attendance-support approaches rather than assuming effectiveness.
- Confirm whether recorded-history data quality is sufficient for operational use.
- Assess whether additional Data Science features materially improve decision quality enough to justify their use.
- Test stakeholder comprehension and usability of the dashboard and decision rules.

Week 7 bridge — Week 6 establishes a validated decision-support framework. Week 7 should determine whether the proposed attendance-support strategy produces measurable improvement before wider deployment.

14. Recommended Executable Notebook Structure

Order	Markdown cell	Code / expected output
1	Title, objective, Week 5→6 transition	Imports + configuration
2	Dataset & analytical scope	Load data + shape + types
3	Data quality / validation	SSOT and population checks
4	KPI revalidation	Measures + KPI reference table
5	AQ1 Booking Lead Time	Bands, CI, Spearman, MW, RR/OR, sensitivity
6	AQ2 Previous No-Shows	0/1/2/3+ grouping, gaps and tests
7	AQ3 Combined Signals	2×2 matrix + additive / interaction models
8	AQ4 Contextual Validation	Reminder + targeted Distance analysis where justified
9	Cross-Track Integration Back	Analytics ↔ DS convergence / nuance table
10	Evidence hierarchy	Final signal classification
11	Decision translation	Structured synthesis; no unnecessary new analysis
12	Measurable recommendations	Action / owner / KPI / test / decision-rule table
13	Limitations & Week 7	Final validation checklist

Code rule — The final notebook should not display every exploratory attempt. Retain only the code required to reproduce, validate and trace the evidence used in the final decision-support narrative.